

# Alessandro Pasqui

## Curriculum Vitae

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### Personal Information

Date of birth 29 - 11 - 1995  
Place of birth Rome, Italy

### Education

Sep 2021 – Oct 2025 (expected) **MSCA PhD Candidate at PSL Université de Paris**   
**Collège de France, Centre Interdisciplinaire de Recherche en Biologie (CIRB)  
& École Normale Supérieure**

Subject: AI-based Methods for Inverse Problems and Computer Vision

Supervisor: Dr. Hervé Turlier

#### **AI-based methods for inverse problems in biological cell systems**

In the first phase of the PhD, developed *ZAugNet*, a self-supervised, machine learning-based method for z-slice augmentation in 3D bio-imaging. This approach integrates concepts from adversarial learning and knowledge distillation. The manuscript is available here: <https://arxiv.org/abs/2503.04843>.

In the second phase, developed *VertAX*, a computational framework for efficient vertex-based modeling and inverse problem solving via bilevel optimisation in confluent tissues. Implemented in JAX, *VertAX* is fully differentiable through automatic differentiation and allows the inference of optimal cellular parameters to reproduce a desired macroscopic behavior or to fit experimental data. A manuscript on this work is currently in preparation.

Sep 2018 – Mar 2021 **Master's Degree at Sapienza University of Rome**   
**Department of Physics  
& Italian Institute of Technology, Center For Life Nano Science (CLNS-IIT)**

Subject: Theoretical Physics (Statistical Mechanics)

Supervisors: Prof. Giancarlo Ruocco, Dr. Giorgio Gosti, Dr. Mattia Miotto

#### **High performance matching algorithm based on orthogonal polynomials**

Developed a computational algorithm to detect complementary-shaped object portions, applied to 2D jigsaw puzzles and 3D archaeological artifacts. The method used local shape descriptors based on orthogonal polynomials. This work contributed to a broader CLNS-IIT project on spike protein–receptor interactions in viral infections (Milanetti et al. 2020).

Grade: 110 (with honors) / 110

Sep 2014 – Jan 2018 **Bachelor's Degree at Sapienza University of Rome**   
**Department of Physics**

Subject: Physics

Supervisor: Prof. Federico Ricci-Tersenghi

**Percolation in models with long-range interactions**

Studied, through theoretical analysis and numerical simulations, the critical point and critical exponents in percolation models involving long-range interactions.

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## Research Experiences

- Sep 2023 – Dec 2023 **Italian Institute of Technology, Center for Life Nano Science (CLNS-IIT)**  
**Rome, Italy**  
***Visiting Researcher***  
Supervisors: Prof. Giancarlo Ruocco, Dr. Giorgio Gosti, Dr. Mattia Miotto  
Visiting period as part of a collaboration to test VertAX, a differentiable vertex-model that I had developed to solve inverse problems on real microscopic data for confluent tissues.
- Apr 2021 – Aug 2021 **Collège de France, Centre Interdisciplinaire de Recherche en Biologie**  
**Paris, France**  
***Research Engineer***  
Supervisors: Dr. Hervé Turlier  
Image-to-image translation via convolutional neural networks for microscopy images in the context of developmental biology: from transmitted light microscopy to fluorescence microscopy (and back).
- Jan 2020 – Feb 2020 **Sapienza University of Rome**  
**Rome, Italy**  
***Intern***  
Supervisor: Dr. Francesca Colaioni  
Connection properties of the directed graph related to the election results for new Wikipedia administrators.
- Mar 2019 – Jul 2019 **Italian Institute of Technology, Center for Life Nano Science (CLNS-IIT)**  
**Rome, Italy**  
***Intern***  
Supervisors: Prof. Giancarlo Ruocco and Dr. Giorgio Gosti  
Pairwise correlation function for neural cells during neural rosettes development through automatic segmentation algorithms based on deep learning using convolutional neural networks.

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## Programming

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|------------------------|-----------|
| <b>JAX/Pytorch</b>     | excellent |
| <b>Python</b>          | excellent |
| <b>LaTeX</b>           | excellent |
| <b>C/C++</b>           | advanced  |
| <b>Shell Scripting</b> | advanced  |

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## Languages

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|----------------|----------|
| <b>English</b> | fluent   |
| <b>French</b>  | fluent   |
| <b>Italian</b> | fluent   |
| <b>Spanish</b> | advanced |

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## Honors and Awards

- Jun 2024 **II EELISA Student Scientific Competition**  
**Budapest, Hungary**  
Awarded *first prize* for outstanding research contribution in the Chemical and Bioengineering category. Featured in multiple interviews highlighting the scientific and societal impact of the project: [\*EELISA interview\*](#), [\*PSL interview\*](#).
- Aug 2021 **Marie Skłodowska-Curie Actions (MSCA) - COFUND Fellowship**  
**AI4theSciences Doctoral Program 2021 at PSL Université Paris**  
**Paris (France)**  
Awarded a *Marie Skłodowska-Curie Actions (MSCA) COFUND Fellowship* as part of a highly competitive European PhD program, recognizing academic excellence and research potential.

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## List of Publications

- 2025 **Self-Supervised Z-Slice Augmentation for 3D Bio-Imaging via Knowledge Distillation**  
Preprint available on arXiv: <https://arxiv.org/abs/2503.04843>
- 2025 (expected) **VertAX: Inverse Vertex-Based Modeling for Tissue Mechanics**  
Differentiable computational framework for efficient vertex-based modeling and inverse problem solving via bilevel optimisation in confluent tissues. Manuscript in preparation.
- 2025 (expected) **AlphaMIC: Synthetic Microscopy Image Generation via CycleGANs**  
In collaboration with the [Turlier Lab](#). Development of a software tool to generate realistic artificial microscopy images using cyclic generative adversarial networks. Manuscript in preparation.
- 2025 (expected) **3D Nuclear Roughness Estimation in Patient-Derived Organoids**  
In collaboration with [Dr. Kate Miroshnikova](#). Analysis of 3D nuclear morphology in patient-derived colonic organoids using spherical harmonics as local shape descriptors for pathological condition classification. Manuscript in preparation.

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## Conferences and Talks

- Feb 2025 **GDR IASIS - Unrolling and un/self/supervised learning for inverse problems**  
**Inria Paris, France**  
Poster presentation.
- Feb 2025 **Les Houches Physics School - From Soft Matter to Biophysics**  
**École de physique des Houches, France**  
Poster presentation.
- Sep 2024 **Institute of Physics - AI for Biological Physics**  
**IoP London, United Kingdom**  
Talk (30 minutes).

Jul 2024 **World Congress on Computational Mechanics (WCCM/PANACM)**  
**Vancouver, Canada**  
Talk (20 minutes).

May 2024 **PSL Doctoral Conference - AI4theSciences**  
**PSL Paris, France**  
Talk (20 minutes).

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## Peer Review

Apr 2023 **21st IEEE International Symposium on Biomedical Imaging (ISBI)**  
**Athens, Greece**  
Served as a peer reviewer for the “Light My Cells: Bright Field to Fluorescence Imaging Challenge,” organized by the French National Research Infrastructure for Biological Imaging (FBI) as part of the ISBI 2024 conference.

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## Organizing Committees

Mar 2023 – Present **CIRB Seminar Series at Collège de France**  
**Paris, France**  
Member of the organizing committee for the weekly seminar series of the Biology Department at Collège de France. Responsibilities include inviting speakers, chairing sessions, and coordinating post-seminar receptions.

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## Other Experience

Jan 2016 – Dec 2016 **Librarian at Sapienza University of Rome, Department of Physics**  
**Rome, Italy**  
Selected for a merit-based internship to support library operations. Responsibilities included assisting patrons, managing check-in/check-out of materials, and classifying/cataloguing books and publications using standard library classification systems.

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## Volunteering

Oct 2024 **French Science Festival at Collège de France**  
**Paris, France**  
Organized and ran a public science outreach stand at the Collège de France as part of the French Science Festival, aimed at making scientific research accessible to a general audience.

Sep 2024 **European Heritage Days at Collège de France**  
**Paris, France**  
Contributed to the organization of events for the European Heritage Days at the Collège de France, helping welcome visitors and explain the historical and scientific significance of the institution.

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## Contacts for References

### **Dr. Hervé Turlier**

Group Leader at Collège de France (CIRB)

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